

What is token?

A token is a unit of text - roughly 4 characters or 0.75 words in English.

what's the capital of India?

Input Tokens: 6

Characters: 28

1,000 words $\sim$ 1,300 Tokens

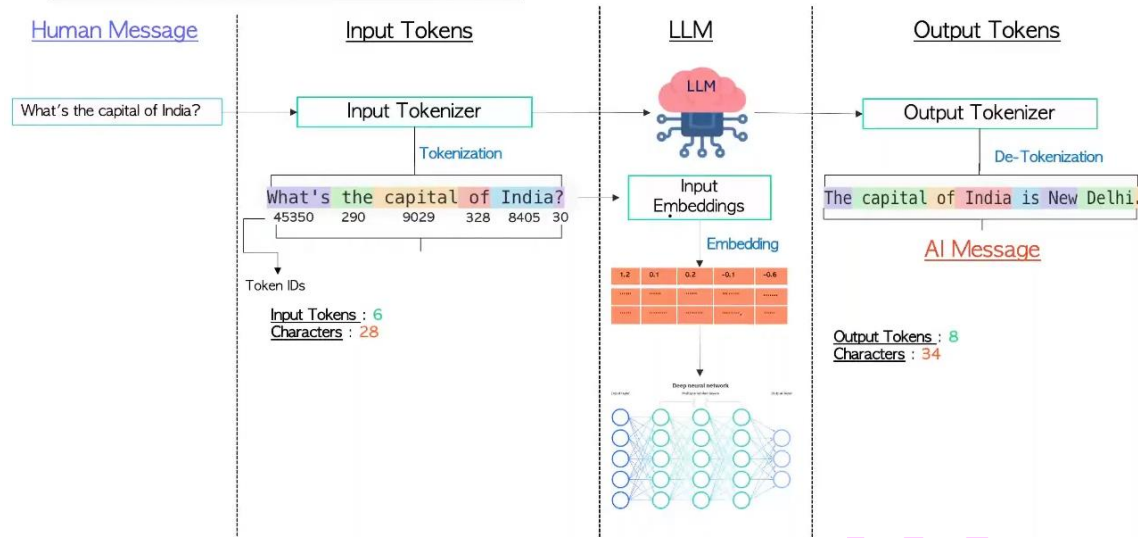
Tokenization (Stage-1)

Tokenization is the process of breaking down text into smaller pieces (called tokens) that a language model can understand and process.



It's required because models like GPT don't understand raw text — they only understand numbers.

How a single message travels through an LLM



Token Pricing

Each token processed by an LLM — whether input or output — contributes to the total cost of using the model.

Token Type	Description
Input Tokens	Tokens sent to the model (your prompt + context)
Output Tokens	Tokens generated by the model (the response) (Costlier)

Model	Input Tokens Price	Output Tokens Price
GPT-5	\$1.25 per 1 M tokens	\$10.00 per 1 M tokens
Grok 4 (by xAI)	\$3.00 per 1 M input tokens	\$15.00 per 1 M output tokens

Total Cost per Request = (Input Tokens × Input Rate) + (Output Tokens × Output Rate)

Example

- You send a prompt that uses 2,000 input tokens.
- The model responds and generates 500 output tokens.

Rates:

- Input = \$1.25 / 1 000 000 tokens → \$0.00000125 per token
- Output = \$10 / 1 000 000 tokens → \$0.00001 per token

Cost calculation:

- Input cost = 2,000 × \$0.00000125 = \$0.0025
- Output cost = 500 × \$0.00001 = \$0.0050
- Total cost for the call = \$0.0075